

Homework 9

Nov 12, 2025

1 (30'). Prove that the Post Correspondence Problem is decidable over the unary alphabet $\Sigma = \{1\}$.

2 (30') Prove that the Post Correspondence Problem is undecidable over the binary alphabet $\Sigma = \{0, 1\}$, assuming that the PCP is undecidable over an arbitrary finite alphabet.

3 (40'). Express the *Twin Prime Conjecture* in first-order Peano Arithmetic using only the symbols $0, S, +, \times, =$ and quantifiers $\forall, \exists$.

- (a) Define a formula $\text{Prime}(x)$ expressing that x is a prime number.
- (b) Define a formula $\text{TwinPrime}(p)$ expressing that p is the smaller number of a twin prime pair.
- (c) Write a single first-order PA formula that states: *for every number x , there exists a larger twin prime $p > x$.*